Nyenje Bay Wind/Waves Analysis Tool

Technical Presentation and Validation Results

Key Informatics

Lake Kariba, Zambia / Zimbabwe

April 2026

Presentation Outline

Project Overview

Location

Nyenje Bay, Lake Kariba
16.53S, 28.83E

Objective

Characterize wind and wave climate for **Floating Photovoltaic (FPV)** design

Data Sources

- ERA5 reanalysis (1971-2020)
- CCMP satellite (debiasing)
- Kariba Airport (validation)
- Local bathymetry survey

Tool Architecture

Layer 4: User Interface
Streamlit Dashboard

Layer 3: Scientific Computing
NumPy, SciPy, Pandas

Layer 2: Data Processing
Xarray, NetCDF, CSV

Layer 1: External Models
SWAN Executable

Nyenje Bay Domain

Domain Parameters

Parameter	Value
Width	4000 m
Height	3500 m
Grid X	9 points
Grid Y	8 points
Total Points	72
Spacing	500 m

Coordinates

Boundary	Value
Center	16.53S, 28.83E
West	28.81113E
East	28.84887E
South	16.54577S
North	16.51423S

Bathymetry Map (CONSTANT)

[Bathymetry Heatmap] Colors: Viridis (dark blue equals deep, yellow equals shallow) Depth range: 1.2 m to 29.0 m Mean depth: 14.8 m

- **CONSTANT** - Same for all SWAN runs
- 9 x 8 grid (72 points)
- Deeper in north (up to 29 m), shallower in south (1.2 m)

Wind Statistics Results

Key Statistics

Parameter	Value
Mean Wind Speed	2.40 m/s
Median Wind Speed	2.21 m/s
Maximum Wind Speed	8.90 m/s
Year of Maximum	1996
Mean Direction	51.6 degrees
Prevailing Direction	**NE**

Interpretation

- Very calm conditions
- Fetch-limited bay
- Ideal for FPV

Wind Averaging Periods

Period	Gust Factor	Calculated
Hourly Mean	1.00	8.90 m/s
3-Minute Average	1.15	10.24 m/s
10-Minute Average	1.00	8.90 m/s
3-Second Gust	1.45	12.91 m/s
10-Second Gust	1.35	12.02 m/s

Example

For maximum wind of 8.90 m/s:

- 3-second gust = 12.91 m/s (ULS design)
- 10-second gust = 12.02 m/s (SLS design)

EWMA Formula for Wind Averaging

Exponential Weighted Moving Average

$$\mathrm{EWMA}_t = \alpha \cdot x_t + (1 - \alpha) \cdot \mathrm{EWMA}_{t-1}$$

$$\alpha = \frac{2}{\mathrm{span} + 1}$$

Parameters

- 3-minute average: span $= 3$, $\alpha = 0.5$
- 10-minute average: span $= 10$, $\alpha = 0.18$

JONSWAP Wave Formula

Dimensionless Fetch

$$\tilde{x} = \frac{g \cdot F}{U^2}$$

Significant Wave Height

$$H_s = 0.0016 \cdot \frac{U^2}{g} \cdot \sqrt{\tilde{x}}$$

Peak Wave Period

$$T_p = 0.286 \cdot \frac{U}{g} \cdot (\tilde{x})^{0.33}$$

Example

For U = 8.90 m/s, F = 4 km:

$$H_s = 0.206 \text{ m } (20.6 \text{ cm})$$

Wave Statistics Results

Parameter	Value
Yearly Maximum Wave Height	0.206 m
Mean Wave Height	0.15 m

All waves less than 25 cm!

- Extremely calm wave conditions
- Fetch-limited (max 4 km)
- Natural wave attenuation

Extreme Value Analysis - Return Periods

Return Period	Wind (m/s)	Wave (m)
2-year	11.2	0.174
5-year	12.8	0.188
10-year	14.0	0.197
25-year	15.6	0.208
50-year	16.8	0.217
100-year	18.0	0.225

Gumbel Distribution

$$x(T) = \mu - \beta \cdot \ln\left(-\ln\left(1 - \frac{1}{T}\right)\right)$$

SWAN Wave Distribution

[Wave Height Distribution Map]
9x8 grid showing wave heights in cm
Colors: RdYlGn (red equals high, green equals low)

- Wave heights vary across domain
- Maximum at northeast quadrant
- Minimum at southwest (sheltered)

SWAN Model Inputs

CONSTANT Input

Bathymetry: Same for all runs

- 9x8 grid from survey data
- Depth range: 1.2 - 29.0 m

VARIABLE Input

Wind: Changes each hour

- From debiased ERA5 data
- u and v components at 72 points
- 438,000 total files (1971-2020)

Validation with Kariba Airport

Location Comparison

Parameter	Difference
Distance	3 km
Elevation	15 m
Surface	Water vs Grass
Fetch	4 km vs Unlimited

Expected Correlation

Due to local effects, r approx 0.40-0.50

- Lake breeze effects
- Topographic sheltering
- Surface roughness differences

Validation Statistics

Metric	Raw ERA5	Debiased	Improvement
Correlation (r)	0.39	0.502	+29%
Bias (m/s)	+2.04	+0.01	99.5%
RMSE (m/s)	2.45	0.52	79%

Validation PASSED!

Time Series Comparison

[Time Series Plot]
Black: Kariba Airport (Ground)
Green: ERA5 Debiased (Nyenje Bay)
Shows good agreement in patterns

Scatter Plot Analysis

[Scatter Plot]
r = 0.502, R-squared = 0.252
Slope close to 1:1

- 25% shared variance (weather)
- 75% local effects (expected)
- Debiasing factor validated

Design Load Cases

Load Case	Wind	Wave	Period
Operational	6.8 m/s	0.10 m	Daily
SLS	7.8 m/s	0.20 m	10-year
ULS	12.91 m/s	0.217 m	50-year
ALS	12.91 m/s	0.225 m	100-year

ULS = Ultimate Limit State
ALS = Accidental Limit State
SLS = Serviceability Limit State

FPV Component Design Values

Component	**Design Value**	**SF**
Mooring lines	12.91 m/s gust	1.5
Anchors	0.225 m wave	1.5
Support structure	12.91 m/s	1.3
Pontoons	0.217 m wave	1.4
Electrical	0.197 m wave	1.2
Freeboard	>0.5 m	—

SF = Safety Factor

Fetch-Limited Advantages

Natural Protection:
Max fetch = 4 km in prevailing direction
Therefore Max 50-year wave = 0.217 m

- No breakwaters required
- Reduced mooring tension
- Lower structural requirements
- **Ideal for FPV installation**

Wind Rose Output

[Wind Rose Diagram]
Prevailing direction: NE (51.6 degrees)
32% from NE, 18% from ENE

- Directional distribution
- Speed bins (0-2, 2-4, 4-6, 6-8 m/s)
- Calm conditions: 8%

Wave Height Distribution Output

[Histogram]
Wave height distribution
Peak at 0.2-0.4 m

- 65%: 0.2-0.4 m
- 28%: 0.4-0.8 m
- 6%: 0.8-1.2 m
- 1%: $>$1.2 m

Return Period Chart Output

[Gumbel Fit Plot]
Return period vs wave height
Log scale on x-axis

- 50-year wave: 0.217 m
- 100-year wave: 0.225 m
- Very stable wave climate

Technology Stack

Component	Technology	Purpose
UI Framework	Streamlit	Web dashboard
Scientific	NumPy	Array operations
Data Analysis	Pandas	DataFrames
NetCDF	Xarray	ERA5 reading
Statistics	SciPy	Gumbel distribution
Visualization	Plotly	Interactive plots
Wave Model	SWAN	Wave distribution

Key Findings Summary

1. **Wind Climate:**
   - Mean: 2.40 m/s from NE
   - Maximum: 8.90 m/s (1996)
   - 50-year gust: 12.91 m/s
2. **Wave Climate:**
   - Yearly maximum: 0.206 m
   - 50-year wave: 0.217 m
   - 100-year wave: 0.225 m
3. **Validation:**
   - $r = 0.502$ with Kariba Airport
   - 99.5% bias reduction
   - Validation PASSED

Design Recommendations

Parameter	50-Year	100-Year
Wind (3-sec gust)	12.91 m/s	12.91 m/s
Significant Wave	0.217 m	0.225 m
Maximum Wave	0.33 m	0.34 m
Freeboard Required	>0.5 m	>0.5 m

Safety Factors

- Mooring/Anchor: **1.5**
- Support structure: **1.3**
- Pontoons: **1.4**

Why Nyenje Bay is Ideal for FPV

Extremely Calm Conditions

- Lowest wind speeds on Lake Kariba
- Mean wind: only 2.40 m/s
- Maximum wave: only 0.225 m (100-year)
- Fetch-limited bay provides natural protection
- **No breakwaters required**
- **Simple, low-cost mooring**

HIGHLY RECOMMENDED FOR FPV DEVELOPMENT

Thank You

Questions?

Key Informatics
Nyenje Bay, Lake Kariba
16.53S, 28.83E

GitHub: https://github.com/chawas/nyenje$_{g}ithub$